



The extract, transform, and load data pipeline pattern

The extract, transform, and load data pipeline pattern

- | | |
|----|---|
| 01 | Extract, transform, and load (ETL) architecture |
| 02 | Google Cloud GUI tools for ETL data pipelines |
| 03 | Batch data processing using Dataproc |
| 04 | Streaming data processing options |
| 05 | Bigtable and data pipelines |



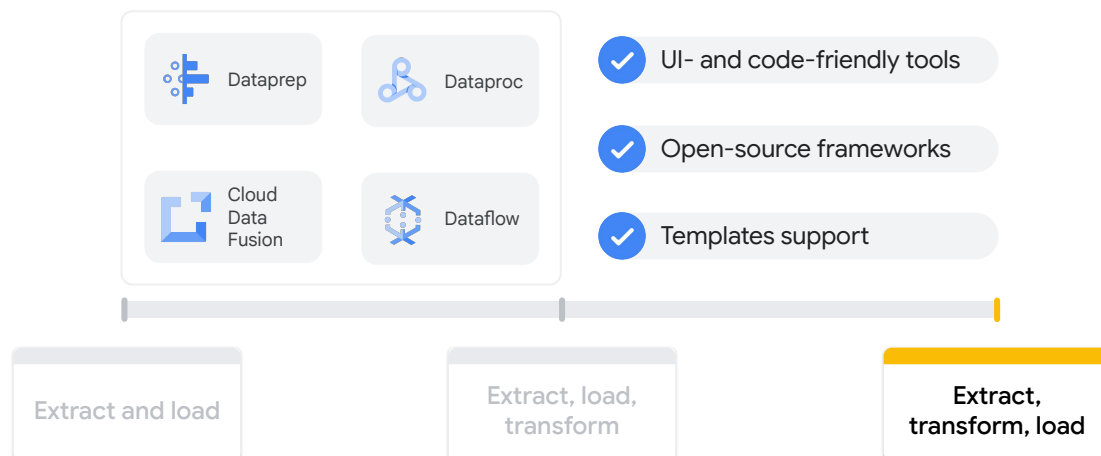
In this module, first, you review the baseline extract, transform, and load architecture diagram. Second, you look at the GUI tools on Google Cloud used for ETL data pipelines. Third, you review batch data processing using Dataproc. Then, you examine using Dataproc Serverless for Spark for ETL. Next, you review streaming data processing options on Google Cloud. Finally, you examine the role Bigtable plays in data pipelines.

The extract, transform, and load data pipeline pattern

- | | |
|----|---|
| 01 | Extract, transform, and load (ETL) architecture |
| 02 | Google Cloud GUI tools for ETL data pipelines |
| 03 | Batch data processing using Dataproc |
| 04 | Streaming data processing options |
| 05 | Bigtable and data pipelines |



Google Cloud provides multiple services for distributed data processing



Google Cloud

The extract, transform, and load data pipeline pattern focuses on data being adjusted or transformed prior to being loaded into BigQuery. Google Cloud offers a variety of services to handle distributed data processing.

For developers who prefer visual interfaces, there are user-friendly tools like Dataprep and Cloud Data Fusion.

If you favor open-source frameworks, Dataproc and Dataflow are an option.

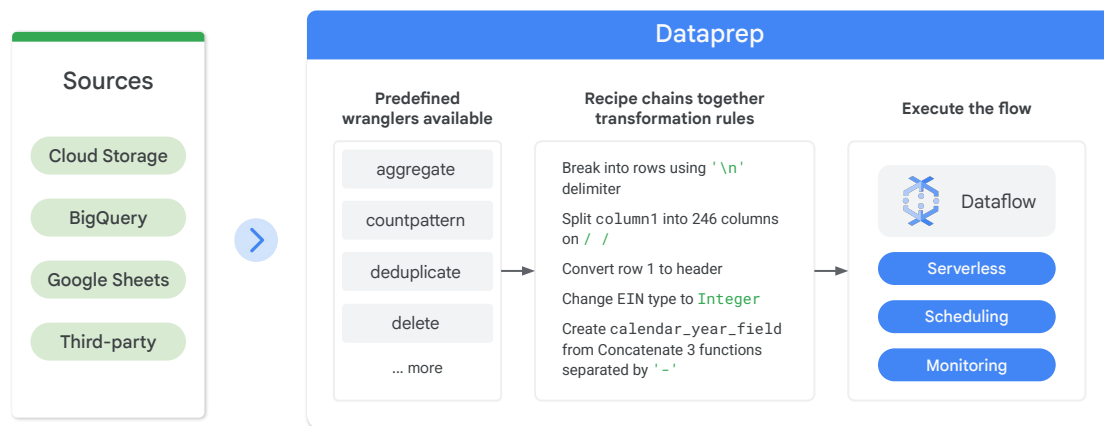
And to streamline your workflows, template support is provided across various stages of data processing, from extraction and loading to full-fledged transformation.

The extract, transform, and load data pipeline pattern

- | | |
|----|---|
| 01 | Extract, transform, and load (ETL) architecture |
| 02 | Google Cloud GUI tools for ETL data pipelines |
| 03 | Batch data processing using Dataproc |
| 04 | Streaming data processing options |
| 05 | Bigtable and data pipelines |



Dataprep by Trifacta is a serverless, no-code solution to build data transformation flows



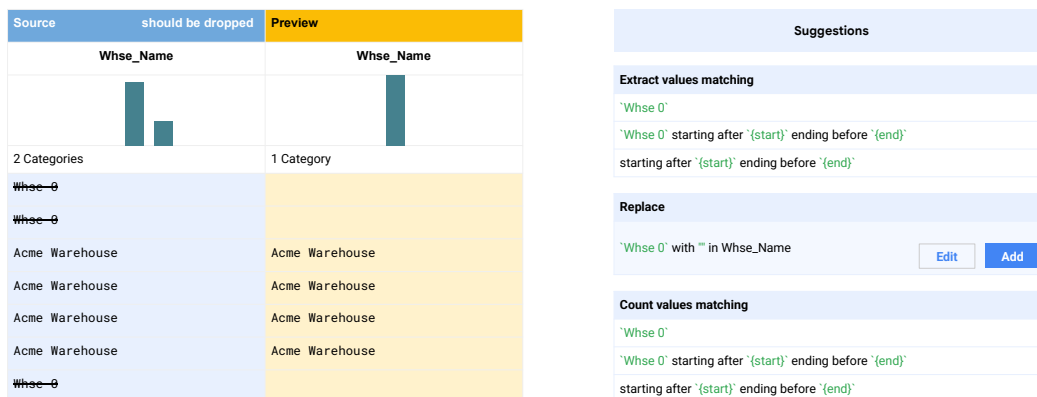
Google Cloud

Dataprep by Trifacta simplifies data transformation by offering a serverless, no-code solution.

It connects to various data sources, provides pre-built transformation functions, and allows users to chain these functions together into recipes.

The resulting transformation flows can be executed seamlessly with Dataflow, while also providing capabilities for scheduling and monitoring.

Dataprep lets you preview your transformations and gives recommendations for your flow



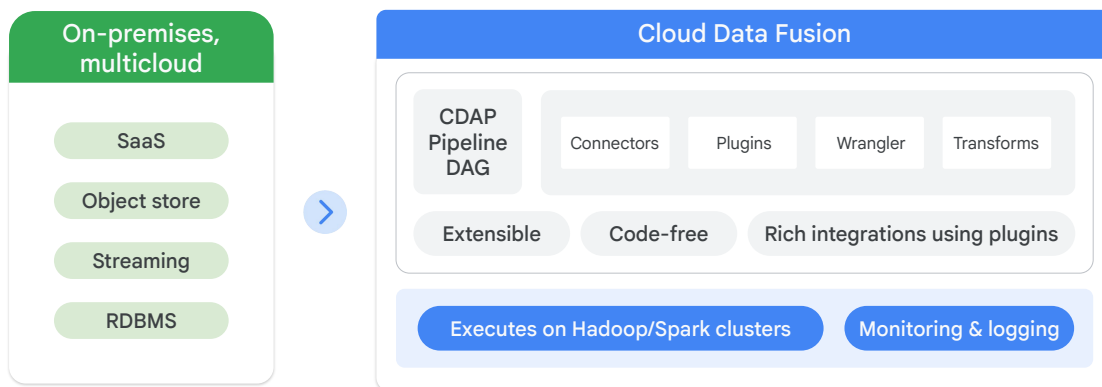
Dataprep provides a visual interface where you can see the impact of your data transformations before applying them.

It also offers intelligent suggestions to help you refine your data cleaning and preparation process, like extracting specific values or replacing patterns within your data.

References:

<https://help.alteryx.com/Dataprep/en/product-overview.html##>

Cloud Data Fusion is a GUI-based enterprise data integration service



Google Cloud

Cloud Data Fusion is a user-friendly, GUI-based tool designed for enterprise data integration.

Cloud Data Fusion seamlessly connects to various data sources, both on-premises and in the cloud.

You can build data pipelines without coding using its drag-and-drop interface and pre-built transformations.

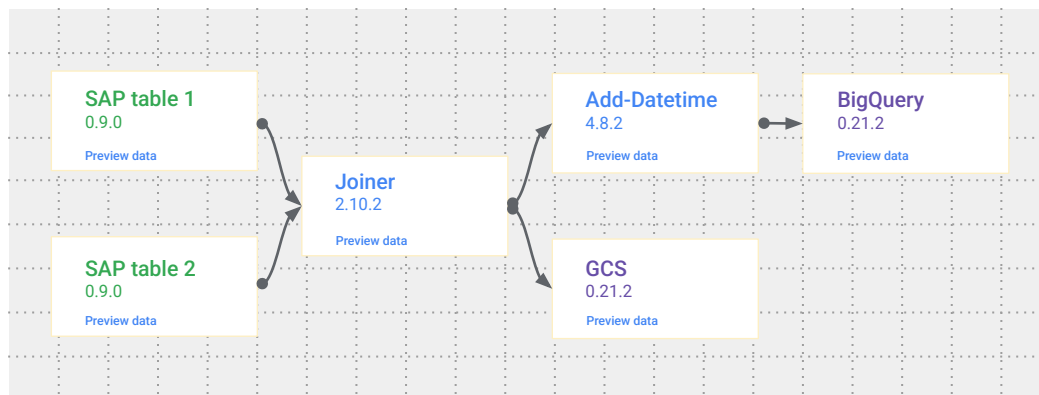
The platform is extensible, allowing for custom plugins, and it executes on powerful Hadoop/Spark clusters for efficient processing.

References:

Huge set of plugins available to integrate data from everywhere:

<https://cloud.google.com/data-fusion/plugins?hl=en>

Create and deploy data integration pipelines in Cloud Data Fusion Studio



Google Cloud

Cloud Data Fusion Studio easily allows the creation of data pipelines with visual tools.

The example pipeline shows two SAP tables being used as data sources. The two tables are joined together and then one outbound leg is written to a Cloud Storage bucket. The other leg undergoes an "Add-Datetime" transformation and is outputted to BigQuery.

The pipeline also highlights the ability to preview data at different stages.

References:

reference for this integration pipeline (SAP>BQ):

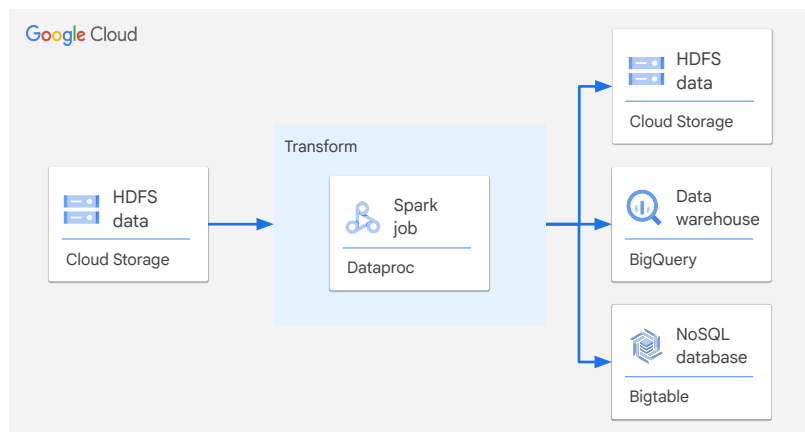
<https://blog.searce.com/sap-to-bigquery-data-ingestion-using-data-fusion-part-2-dde072d2cf6f>

The extract, transform, and load data pipeline pattern

- | | |
|----|--|
| 01 | Extract, transform, and load (ETL) architecture |
| 02 | Google Cloud GUI tools for ETL data pipelines |
| 03 | Batch data processing using Dataproc |
| 04 | Streaming data processing options |
| 05 | Bigtable and data pipelines |



Run your Apache Hadoop and Spark workloads on Dataproc



Google Cloud

Dataproc allows you to seamlessly run your Apache Hadoop and Spark workloads on Google Cloud.

You can leverage HDFS data stored on Cloud Storage, and use Dataproc to perform transformations with Spark jobs.

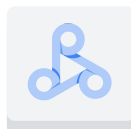
The results can then be easily stored in various destinations like Cloud Storage, BigQuery, or NoSQL databases like Bigtable, all within the Google Cloud ecosystem.

Dataproc is the managed service for data processing with Hadoop and Spark

Runtimes for
Compute Engine, GKE,
Serverless for Spark

Workflow templates
with dependencies

Integration with other
storage services for
HDFS replacement



Dataproc

Rich open-source
ecosystem (Hadoop, Spark,
Pig, Hive pre-installed)

Reactive **autoscaling**
policies

Permanent and
ephemeral clusters with
scheduled deletion

Google Cloud

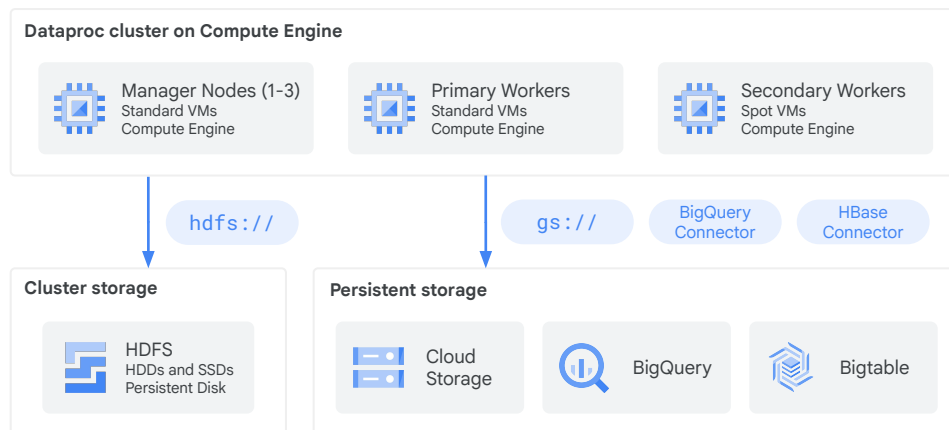
Dataproc is Google Cloud's managed service for data processing using Hadoop and Spark.

It offers flexibility with runtimes on Compute Engine, GKE, and Serverless Spark, and provides a rich open-source ecosystem.

Dataproc simplifies cluster management with workflow templates, autoscaling, and the option for both permanent and ephemeral clusters.

It also integrates seamlessly with other Google Cloud storage services, eliminating the need for disk-based HDFS.

Dataproc clusters can use HDFS on persistent disks or other Google Cloud storage services



Google Cloud

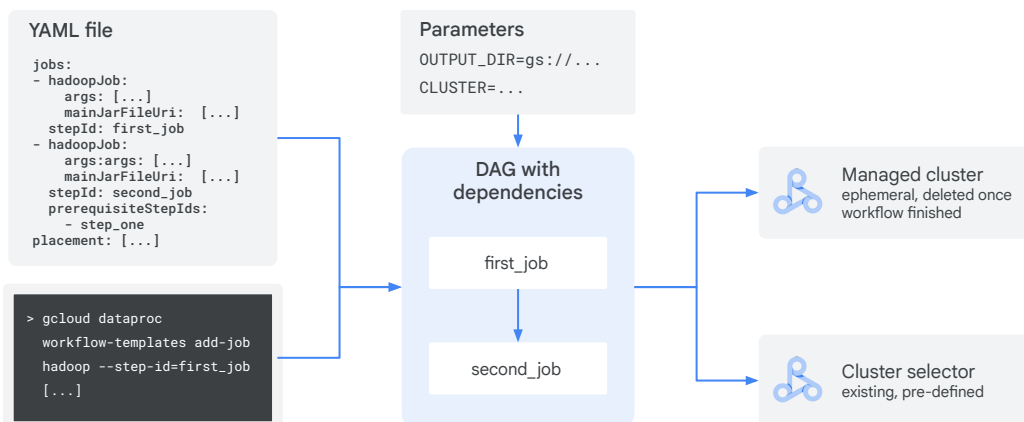
Dataproc clusters on Compute Engine offer flexible storage options.

Clusters can utilize HDFS on persistent disks for cluster storage, or leverage other Google Cloud storage services like Cloud Storage for persistent data.

Additionally, Dataproc integrates with BigQuery and Bigtable using connectors, enabling seamless interaction with these data stores.

This setup allows users to choose the most suitable storage solution for their specific needs while taking advantage of Dataproc's processing capabilities.

Manage and execute workflows with dependencies using Dataproc Workflow Templates



Google Cloud

Dataproc Workflow Templates allow you to define and manage complex data processing workflows with dependencies between jobs.

You can specify these workflows in a YAML file, providing details about the jobs like Hadoop or Spark, their order of execution, and any required parameters.

These templates can then be submitted to Google Cloud using the 'gcloud' command-line tool, where they will be executed on either a managed, ephemeral cluster or an existing, pre-defined cluster.

References:

Two ways to create workflow templates:

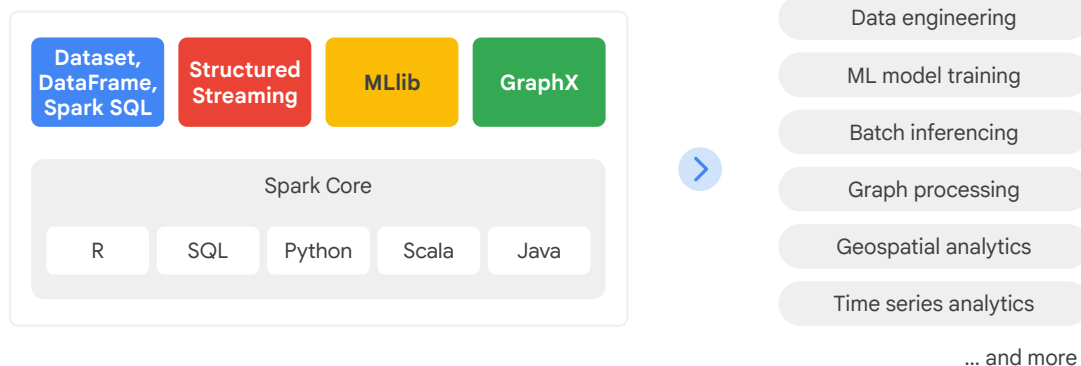
YAML: <https://cloud.google.com/dataproc/docs/concepts/workflows/use-yamls>

gcloud: <https://cloud.google.com/dataproc/docs/concepts/workflows/use-workflows>

Parameters:

<https://cloud.google.com/dataproc/docs/concepts/workflows/workflow-parameters>

Apache Spark is a popular framework for many data processing tasks

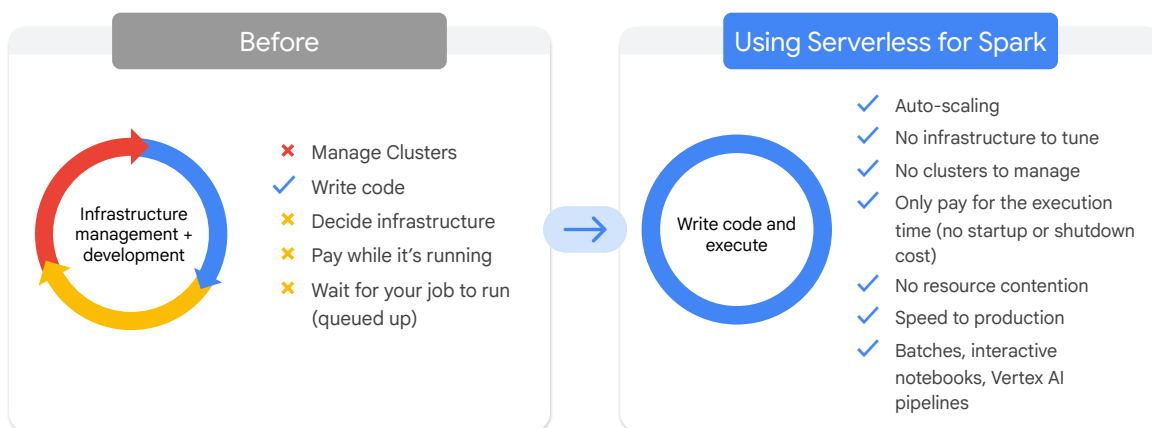


Apache Spark is a versatile framework for data processing, offering various capabilities through its components like Spark SQL for structured data, Spark Streaming for real-time data, MLlib for machine learning, and GraphX for graph processing.

Spark supports multiple languages including R, SQL, Python, Scala, and Java, making it accessible to a wide range of users.

With these features, Spark excels in tasks like data engineering, machine learning, analytics, and many more.

Run Spark workloads without cluster management using Dataproc Serverless for Spark



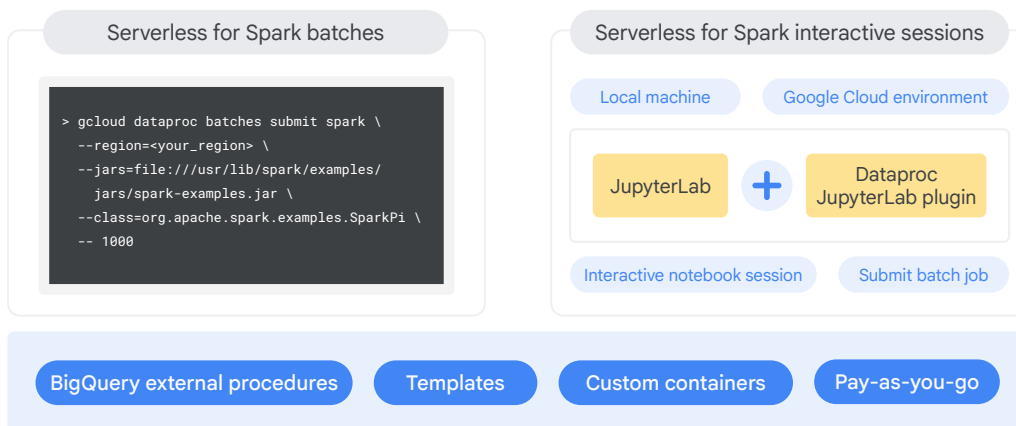
Google Cloud

Dataproc Serverless for Spark simplifies Spark workload execution by eliminating cluster management.

It offers automatic scaling, cost efficiency with pay-per-execution pricing, faster deployment, and no resource contention.

Users can focus solely on writing and executing their code, making it ideal for various Spark use cases like batch processing, interactive notebooks, and Vertex AI pipelines.

Dataproц Serverless for Spark provides batches and interactive notebook session options



Google Cloud

Dataproц Serverless for Spark offers two main execution modes: Serverless for batches and Serverless for interactive notebook sessions.

Batches are submitted using the 'gcloud' command-line tool and are ideal for automated or scheduled jobs.

Interactive sessions leverage JupyterLab, either locally or within the Google Cloud environment, for interactive development and exploration.

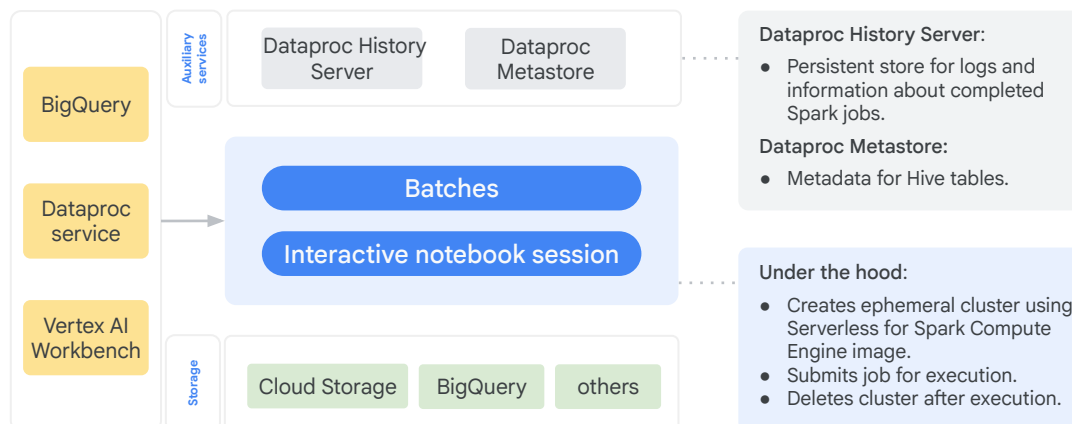
The platform also supports features like BigQuery external procedures, templates, custom containers, and a pay-as-you-go pricing model.

References:

<https://cloud.google.com/dataproц-serverless/docs/quickstarts/spark-batch>

<https://cloud.google.com/dataproц-serverless/docs/quickstarts/jupyterlab-sessions>

Dataproc Serverless for Spark provides rich integrations with Google Cloud services



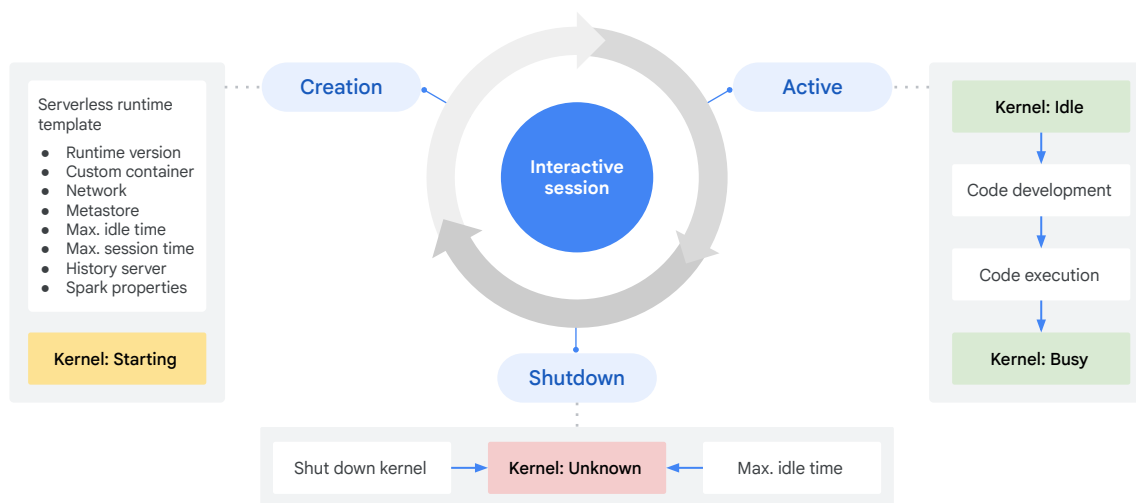
Dataproc Serverless for Spark seamlessly integrates with various Google Cloud services, enhancing its functionality and usability.

It leverages Dataproc History Server and Dataproc Metastore for persistent storage and metadata management.

It interacts with BigQuery for data warehousing and analytics, and with Vertex AI Workbench for machine learning tasks.

Additionally, it utilizes Cloud Storage and other storage services for data storage and retrieval. Behind the scenes, it creates and manages ephemeral clusters for efficient job execution.

The lifecycle of an interactive notebook session



The lifecycle of an interactive notebook session begins with its creation, where various configurations like runtime version and network settings are defined.

Once active, the session allows for code development and execution, with the kernel transitioning between idle and busy states.

The session eventually reaches a shutdown phase, either manually triggered or due to inactivity, leading to the kernel being shut down and its state becoming unknown.

References:

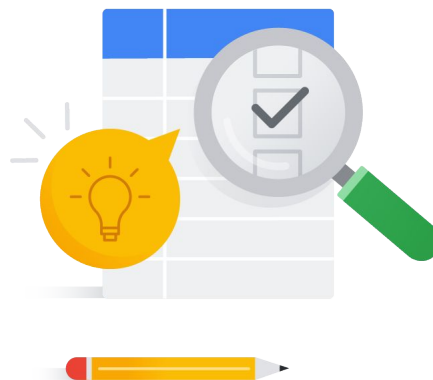
<https://cloud.google.com/dataproc-serverless/docs/quickstarts/jupyterlab-sessions>

Lab (optional): Use Dataproc Serverless for Spark to load BigQuery

🕒 30 min

Learning objectives

- Configure the environment.
- Download lab assets.
- Configure and execute the Spark code.
- View data in BigQuery.



Google Cloud

In this lab, you use Dataproc Serverless for Spark to load BigQuery.

First, you configure the environment.

Next you download lab assets.

You then configure and execute the Spark code.

Finally, you view the data in BigQuery.

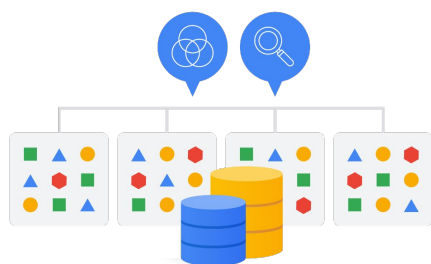
reference: <https://www.cloudskillsboost.google/authoring/labs/31670>

The extract, transform, and load data pipeline pattern

- | | |
|----|---|
| 01 | Extract, transform, and load (ETL) architecture |
| 02 | Google Cloud GUI tools for ETL data pipelines |
| 03 | Batch data processing using Dataproc |
| 04 | Streaming data processing options |
| 05 | Bigtable and data pipelines |



Batch processing versus streaming data processing

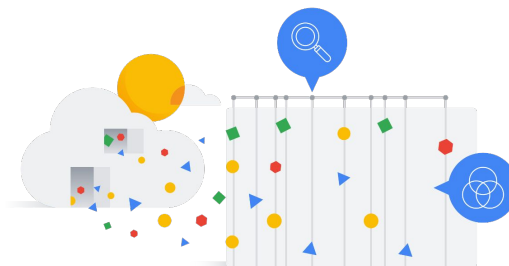


Batch processing

Processing and analysis happens on a set of stored data.

Payroll system

Billing system



Streaming data processing

A flow of data records generated by various data sources.

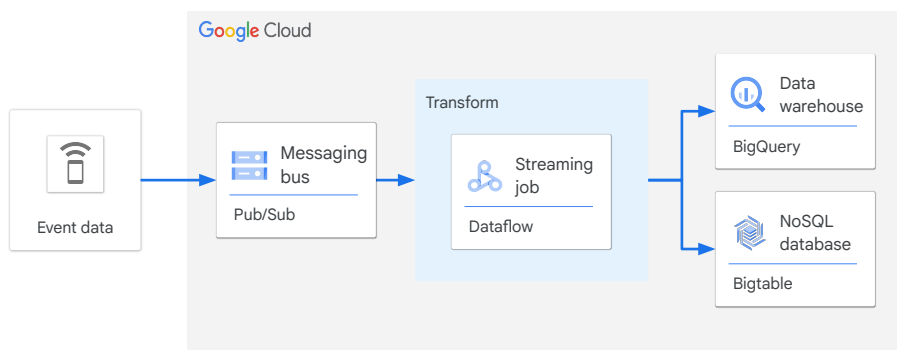
Fraud detection

Intrusion detection

Batch processing involves analyzing a fixed set of stored data, suitable for tasks like payroll or billing systems.

On the other hand, streaming data processing handles a continuous flow of data from various sources, making it ideal for real-time applications like fraud or intrusion detection.

Streaming ETL workloads require continuous data ingestion, processing, and near real-time analytics

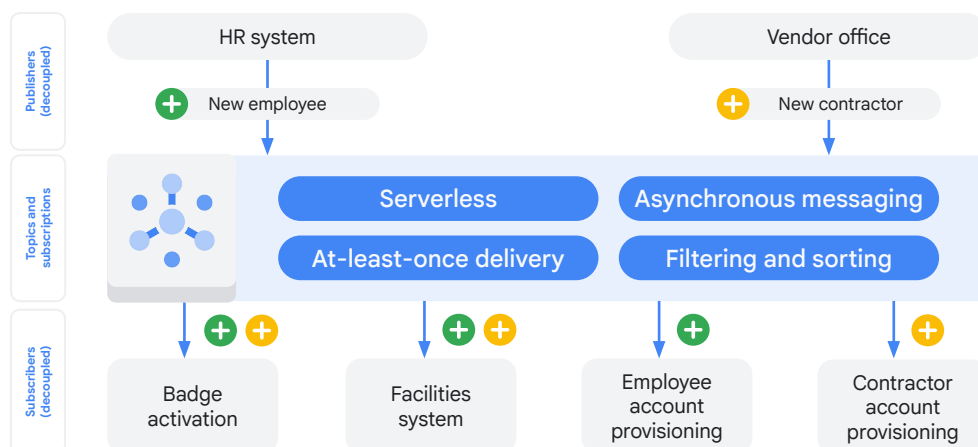


Streaming ETL workloads on Google Cloud involve the continuous ingestion of event data, often through Pub/Sub.

This data is often processed in real-time using Dataflow, allowing for transformations and enrichment.

Finally, the processed data is loaded into various destinations like BigQuery for analytics, enabling near real-time insights, or Bigtable for NoSQL storage.

Use Pub/Sub to ingest high volumes of event data and distribute to different consuming systems



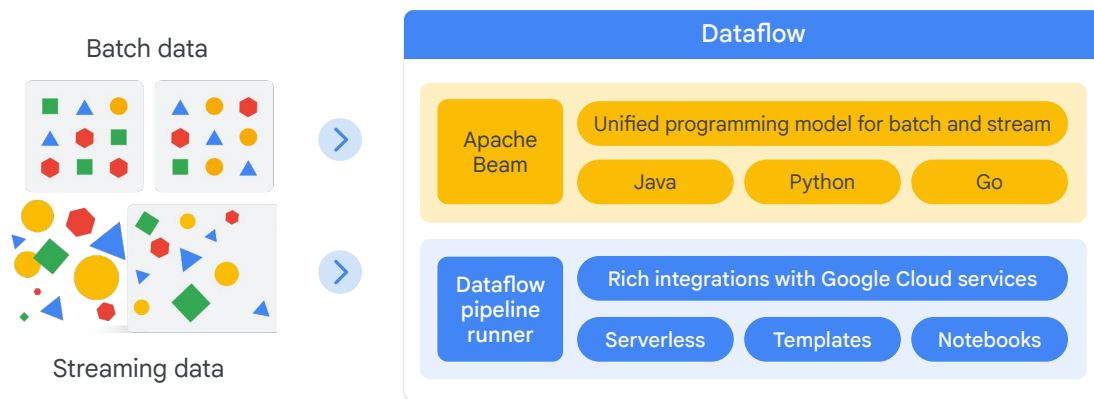
Google Cloud

Pub/Sub can efficiently manage high volumes of event data.

Pub/Sub acts as a central hub, receiving events like "New employee" or "New contractor" from various sources.

Pub/Sub then distributes these events to relevant systems like badge activation, facilities, and account provisioning, ensuring reliable delivery and enabling decoupled, asynchronous communication between systems.

Dataflow can process both batch and streaming data using the Apache Beam programming model



Dataflow leverages the Apache Beam programming framework to efficiently process both batch and streaming data.

This unified approach simplifies development, allowing you to use languages like Java, Python, or Go.

Dataflow seamlessly integrates with other Google Cloud services and offers features like a pipeline runner, serverless execution, templates, and notebooks for a streamlined experience.

Example: Streaming and transforming messages from Pub/Sub to BigQuery

```
with beam.Pipeline(options=pipeline_options) as pipeline:

    # Read messages from Pub/Sub
    messages = pipeline | 'read' >> ReadFromPubSub(topic='<PUBSUB_TOPIC>')

    # Parse messages
    def parse_message(message):
        return json.loads(message.decode('utf-8'))

    parsed_messages = messages | 'parse' >> beam.Map(parse_message)

    # Write parsed messages to BigQuery
    parsed_messages | 'write' >> WriteToBigQuery(
        table='<BIGQUERY_TABLE>',
        schema=table_schema,
        create_disposition=BigQueryDisposition.CREATE_IF_NEEDED,
        write_disposition=BigQueryDisposition.WRITE_APPEND
    )

    # Run the pipeline
    pipeline.run()
```

ReadFromPubSub():

- Retrieves messages from a Pub/Sub topic or subscription.

beam.Map():

- Applies a specified transformation to the message.

WriteToBigQuery():

- Writes the transformed message into a BigQuery table.

This code example demonstrates how to use Apache Beam to stream messages from Pub/Sub, transform them using a parsing function, and then write the results into BigQuery.

The “**ReadFromPubSub**” function retrieves messages, “**Beam.Map()**” applies the parsing transformation, and “**WriteToBigQuery**” loads the transformed data into a specified BigQuery table, creating the table if necessary and appending new data to it.

Use Dataflow templates for recurring pipeline executions with different parameters

+ CREATE JOB FROM TEMPLATE

Dataflow template
Cloud Storage Text to BigQuery (Stream) ▼

Source
gs:// Cloud Storage Input File(s) **BROWSE**

Target
BigQuery output table **BROWSE**

Required Parameters ▼

Benefits of using templates

- Pipeline design is separate from deployment.
- Parameters customize the pipeline and make it reusable with different inputs.
- Templated pipelines can be deployed from the Google Cloud console, command-line interface, or API.
- Use Google-provided pre-built templates for common scenarios.



You can create your own custom templates.

Google Cloud

Dataflow templates allow you to create reusable pipelines for recurring tasks.

You can separate the pipeline design from its deployment, making it easier to manage and update.

By using parameters, you can customize the pipeline for different inputs, increasing its versatility.

These templated pipelines can be easily deployed through various methods, and Google provides pre-built templates for common scenarios.

References:

<https://cloud.google.com/dataflow/docs/concepts/dataflow-templates>

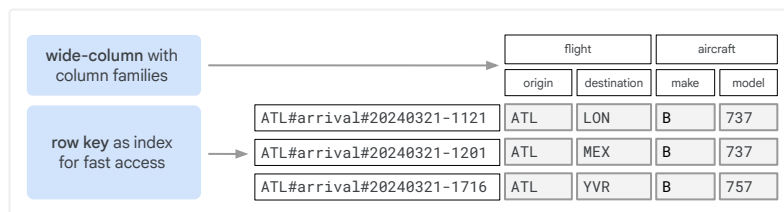
<https://cloud.google.com/dataflow/docs/guides/templates/provided/text-to-bigquery-stream> (used in lab)

The extract, transform, and load data pipeline pattern

- | | |
|----|---|
| 01 | Extract, transform, and load (ETL) architecture |
| 02 | Google Cloud GUI tools for ETL data pipelines |
| 03 | Batch data processing using Dataproc |
| 04 | Streaming data processing options |
| 05 | Bigtable and data pipelines |



Use Bigtable as a sink in your streaming data pipeline for millisecond latency analytics



- ✓ NoSQL queries over large datasets (petabytes)
- ✓ High throughput with millisecond latency
- ✓ Uses open-source HBase API for integration

```
// first row output
ATL#arrival#20240321-1121 column=flight:origin, timestamp: 1711020073, value=ATL
ATL#arrival#20240321-1121 column=flight:destination, timestamp: 1711020073, value=LON
ATL#arrival#20240321-1121 column=aircraft:make, timestamp: 1711020073, value=B
ATL#arrival#20240321-1121 column=aircraft:model, timestamp: 1711020073, value=737
```

Applications:

- Time-series data
- IoT
- Financial data
- Machine learning

Google Cloud

Bigtable is an excellent choice for handling streaming data pipelines that require millisecond-level latency analytics.

Bigtable utilizes a wide-column data model with column families, allowing for flexible schema design.

Row keys serve as efficient indexes for quick data access.

Bigtable's high throughput and low latency capabilities make it suitable for applications like time-series data, IoT, financial data, and machine learning, especially when dealing with large datasets.

References:

Bigtable column and row limits:

<https://cloud.google.com/bigtable/quotas#limits-data-size>

Compare ETL processing options

	Dataprep	Cloud Data Fusion	Dataproc	Dataflow
Applications	Data wrangling	Data integration	ETL workloads	ETL workloads
Integrations	Cloud Storage, BigQuery, others	hybrid, multicloud environments	Cloud Storage, BigQuery, other OSS	Pub/Sub, Cloud Storage, BigQuery, other OSS
Open source	–	CDAP	Apache Hadoop, Spark, other OSS	Apache Beam
Velocity	Batch	Batch, stream	Batch, stream	Batch, stream (recommended)
Serverless	yes	no	no (except Serverless Spark)	yes

Google Cloud

In summary, Google Cloud provides various services for ETL processing.

Dataprep is ideal for data wrangling tasks and offers a serverless option.

Cloud Data Fusion excels at data integration, particularly in hybrid and multi-cloud environments, utilizing the open-source CDAP framework.

Dataproc handles ETL workloads with support for Hadoop, Spark, and other open-source tools, with Serverless Spark as a serverless option.

Lastly, Dataflow, built on Apache Beam, is recommended for both batch and streaming ETL workloads and provides a serverless architecture.

References:

Dataprep connection types:

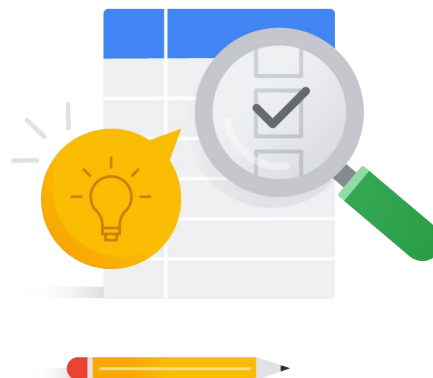
<https://help.alteryx.com/Dataprep/en/platform/connections/connection-types.html>

Lab: Creating a Streaming Data Pipeline for a Real-Time Dashboard with Dataflow

🕒 45 min

Learning objectives

- Create a Dataflow job from a template.
- Stream data via Dataflow pipeline into BigQuery.
- Monitor a Dataflow pipeline in BigQuery.
- Analyze results with SQL.
- Visualize key metrics in Looker Studio.



Google Cloud

In this lab, you create a streaming data pipeline for a real-time dashboard with Dataflow.

You create a Dataflow job from a template.

You then monitor a pipeline loading data into BigQuery.

After that, you examine the data loaded using SQL.

Finally, you visualize key metrics using Looker Studio.

reference: https://www.cloudskillsboost.google/catalog_lab/1796

